

Basic Principles of Medicine 1

Module: Foundations to Medicine

Lecture 3

Homeostasis

Dr Stephan Bandelow

sbandelo@sgu.edu



Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Suggestions

- **Meet the Prof**
 - make use of office hours
- **Do MCQs**
 - practice makes perfect
- **Pre-read lecture**
 - get an overview of topic flow
 - highlight terms you don't understand
 - look these up: Google scholar, Pubmed, Wikipedia for overview

Lecture Objectives

SOM.MK.II.BPM1.1.FTM.3.PHYS.0001 Apply the concept of a homeostatic feedback loop to the regulation of key controlled variables (temperature, blood pressure, water content in extra- and intracellular compartments, pH) in the human body.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0002 List the typical values and normal ranges for plasma Na^+ , K^+ , H^+ (pH), HCO_3^- , Cl^- , Ca^{+2} , and glucose, and the typical intracellular pH and concentrations of Na^+ , K^+ , Cl^- , Ca^{+2} , and HCO_3 .

SOM.MK.II.BPM1.1.FTM.3.PHYS.0003 Define the terms “steady state” and “equilibrium” and evaluate if example systems like specific ion and water distributions across a membrane are in either state.

Recommended Reading

Rhoades & Bell, Medical Physiology, 6th edition. Lippincott.
Main Physiology textbook for year 1, ebook version is available via the library.

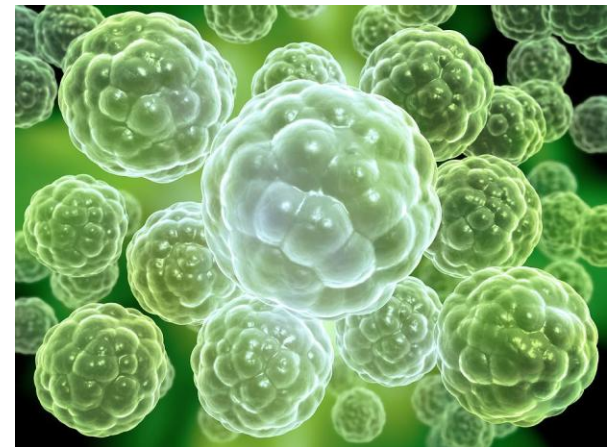
- Chapter 1: Section 'Scope of Physiology'
- Chapter 2: Section 'Basis of physiologic regulation'
- Chapter 23: Sections 'Fluid compartments' and 'Water balance'

Outline

- Homeostasis - internal and external environments
- Water distribution in different compartments
- Steady state vs Equilibrium
- Homeostatic mechanisms
 - Positive & negative feedback
 - Glucose and Blood pressure
 - Feed forward
 - Set point
- Homeostatic hierarchy

Homeostasis

- Single cell organisms live within the environment and their physiology/biochemistry is controlled by that environment
- Multi-cellular organisms generate their own environment in which the cells live and that environment must be controlled and maintained by the organism

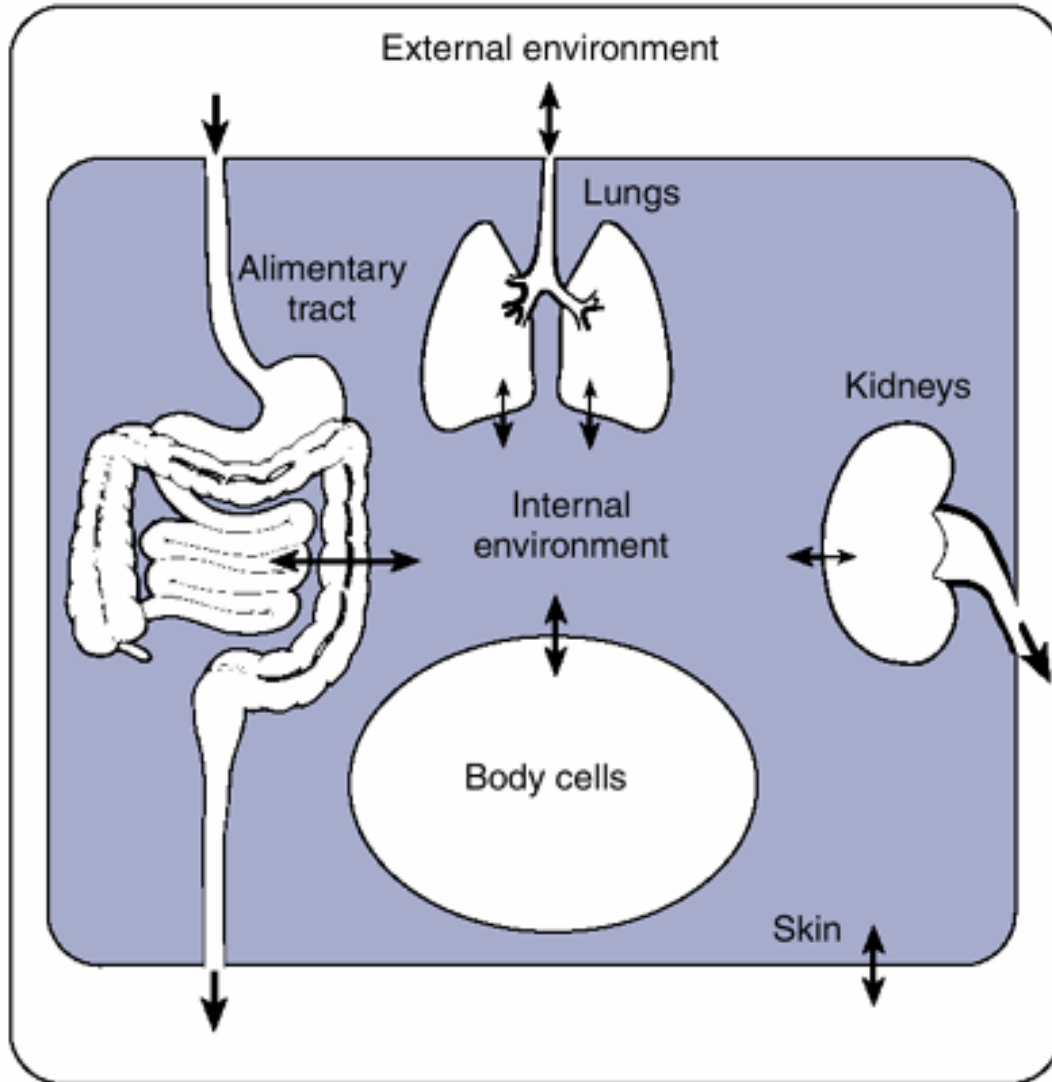


Environments

Variable	Outside	Inside
Temperature	-10 and +40 °C	37 °C
PO ₂	160 mmHg	PaO ₂ 95 mmHg
PCO ₂	0.23 mmHg	40 mmHg
Water	0 – 90 g/kg of air	600 g/kg tissue
pH	?/variable	pH 7.4

Milieu Interieur: Environment must be maintained within limits for optimal biological function: gas tensions, partial pressure of oxygen and carbon dioxide, glucose concentration, osmotic pressure, ion concentrations, pH, temperature.

External and Internal Environments



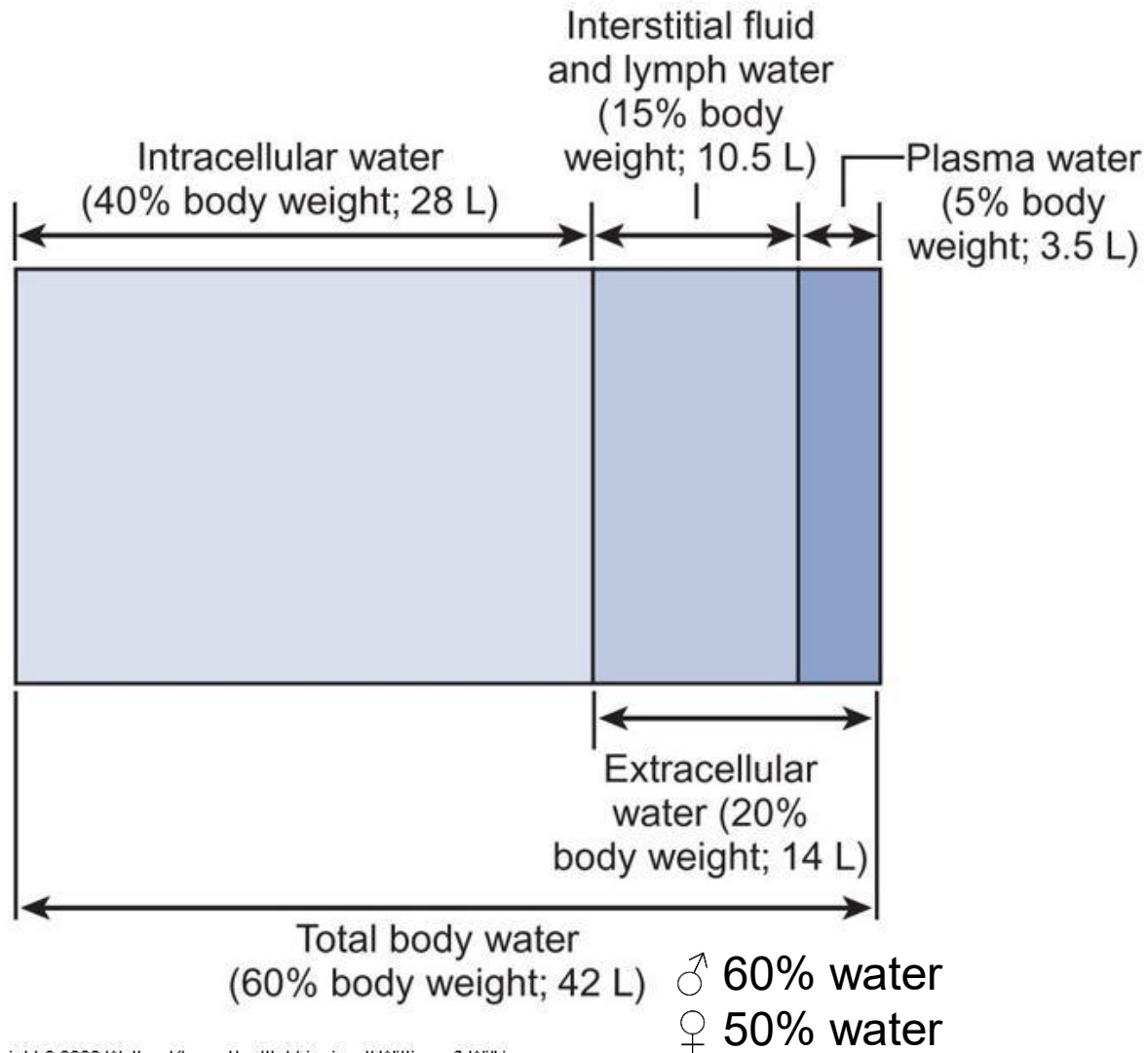
By controlling its milieu, the organism is no longer at the mercy of external environment variances

The homeostatic mechanisms evolved to maintain a steady state within our internal environment in the face of these external changes

Homeostasis

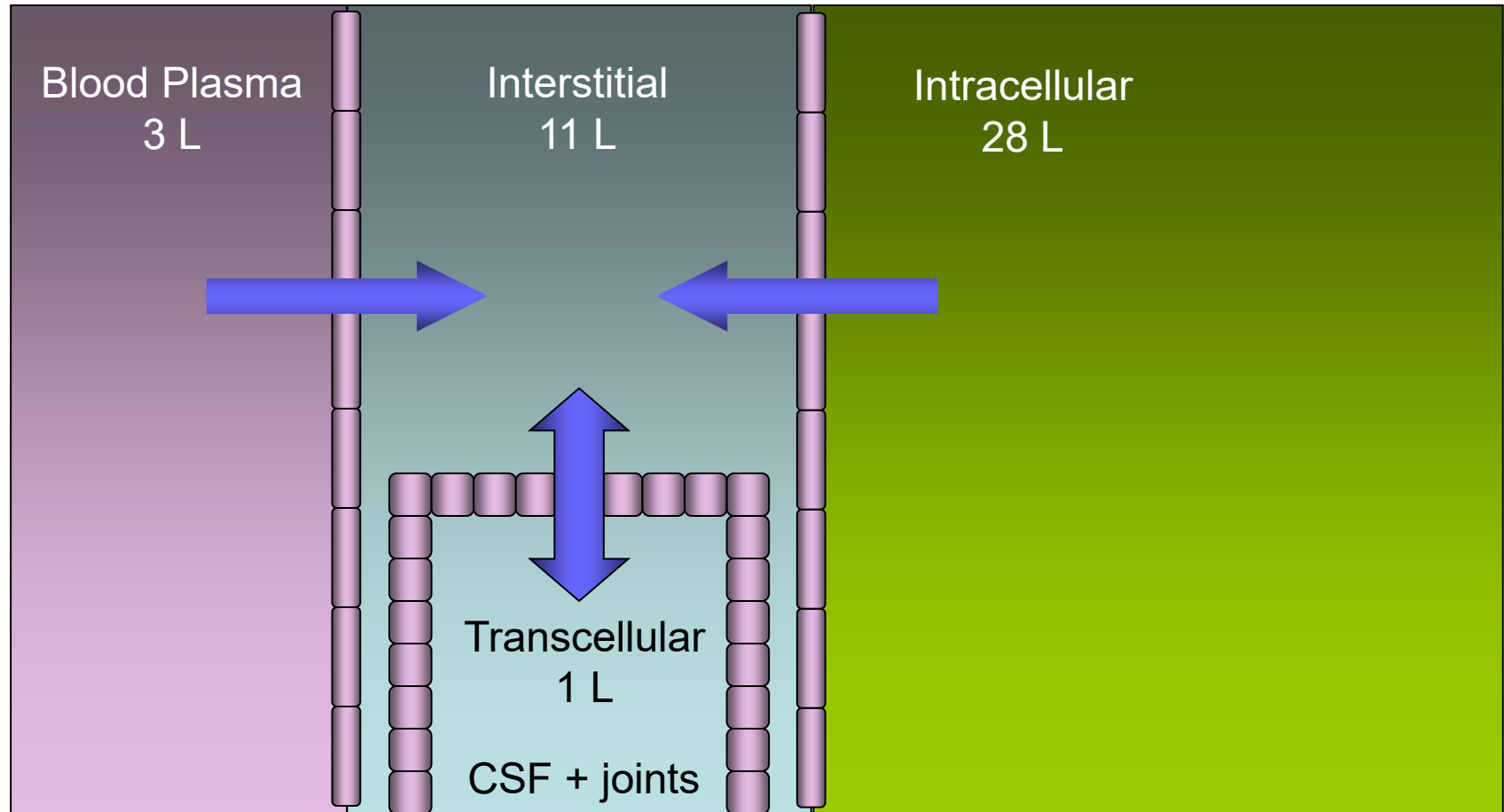
- External environment is out of our control
- In our tissues, the intracellular fluid (ICF) within the cells communicates with the extracellular fluid (ECF) that bathes and surrounds the cells; fluids must be regulated for cells to function normally
- Cells evolved to function optimally within a narrow set of environmental parameters – temperature, pH, oxygen tension - vital that internal environment be kept stable → maintain a dynamic equilibrium
- Cell requirements: Maintain its functional integrity; act with other cells to perform required organ function; contribute to organism survival
- **Homeostasis**: Ability of the body to maintain a relatively stable internal environment despite external variances
- **Physiology**: The study of the various mechanisms that maintain homeostasis

Water Distribution



Copyright © 2009 Wolters Kluwer Health | Lippincott Williams & Wilkins

Compartments and their Relationship



1/3 of total body water is Extracellular Fluid (ECF): Fluid that bathes the cells. Plasma; Interstitial fluid and specialized transcellular fluid compartments – cerebrospinal, synovial, and pleural fluids

2/3 of total body water is Intracellular Fluid (ICF): Fluid within the cells. Substances in plasma must pass through the interstitial fluid before entering the cells. Water distributes between compartments depending on the solute concentration present in each compartment

Ionic Composition of ECF and ICF

Substance and Units	Extracellular Fluid	Intracellular Fluid*
Na ⁺ (mmol/L)	140	14
K ⁺ (mmol/L)	4	120
Ca ⁺² (mmol/L)	2.5	1 × 10 ⁻⁴
Mg ⁺²	0.8	20
Cl ⁻ (mmol/L)	105	10
HCO ₃ ⁻ (mmol/L)	24	10
pH*	7.4	7.1
Osmolarity (mOsm/L)	290	290

*pH is $-\log_{10}$ of the $[H^+]$; pH 7.4 corresponds to $[H^+]$ of 40×10^{-9} Eq/L.

Body Fluid Constituents

Plasma	Interstitial Fluid	Intracellular Fluid																											
<table><tr><td rowspan="3">Na⁺ 142</td><td>HCO₃⁻ 24</td></tr><tr><td>Cl⁻ 105</td></tr><tr><td>Protein 15</td></tr><tr><td>K⁺ 5</td><td>HPO₄⁻ 5</td></tr><tr><td>Ca⁺⁺ 5</td><td>SO₄⁻ 4</td></tr><tr><td>Mg⁺⁺ 3</td><td>R⁻ 2</td></tr></table>	Na ⁺ 142	HCO ₃ ⁻ 24	Cl ⁻ 105	Protein 15	K ⁺ 5	HPO ₄ ⁻ 5	Ca ⁺⁺ 5	SO ₄ ⁻ 4	Mg ⁺⁺ 3	R ⁻ 2	<table><tr><td rowspan="2">Na⁺ 144</td><td>HCO₃⁻ 27</td></tr><tr><td>Cl⁻ 118</td></tr><tr><td>K⁺ 5</td><td>HPO₄⁻ 5</td></tr><tr><td>Ca⁺⁺ 5</td><td>SO₄⁻ 4</td></tr><tr><td>Mg⁺⁺ 3</td><td>R⁻ 2</td></tr></table>	Na ⁺ 144	HCO ₃ ⁻ 27	Cl ⁻ 118	K ⁺ 5	HPO ₄ ⁻ 5	Ca ⁺⁺ 5	SO ₄ ⁻ 4	Mg ⁺⁺ 3	R ⁻ 2	<table><tr><td>Na⁺ 6</td><td>HCO₃⁻ 24</td></tr><tr><td rowspan="3">K⁺ 154</td><td>SO₄⁻ 17</td></tr><tr><td>HPO₄⁻ 106</td></tr><tr><td>R⁻ 4</td></tr><tr><td>Mg⁺⁺ 3</td><td>Protein 15</td></tr></table>	Na ⁺ 6	HCO ₃ ⁻ 24	K ⁺ 154	SO ₄ ⁻ 17	HPO ₄ ⁻ 106	R ⁻ 4	Mg ⁺⁺ 3	Protein 15
Na ⁺ 142		HCO ₃ ⁻ 24																											
		Cl ⁻ 105																											
	Protein 15																												
K ⁺ 5	HPO ₄ ⁻ 5																												
Ca ⁺⁺ 5	SO ₄ ⁻ 4																												
Mg ⁺⁺ 3	R ⁻ 2																												
Na ⁺ 144	HCO ₃ ⁻ 27																												
	Cl ⁻ 118																												
K ⁺ 5	HPO ₄ ⁻ 5																												
Ca ⁺⁺ 5	SO ₄ ⁻ 4																												
Mg ⁺⁺ 3	R ⁻ 2																												
Na ⁺ 6	HCO ₃ ⁻ 24																												
K ⁺ 154	SO ₄ ⁻ 17																												
	HPO ₄ ⁻ 106																												
	R ⁻ 4																												
Mg ⁺⁺ 3	Protein 15																												
Osmolarity 290 mOsm	Osmolarity 290 mOsm	Osmolarity 290 mOsm																											

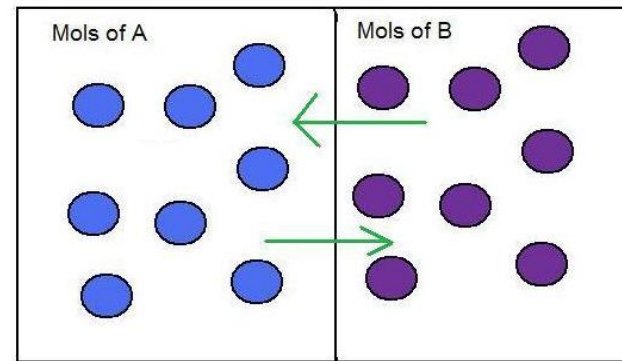
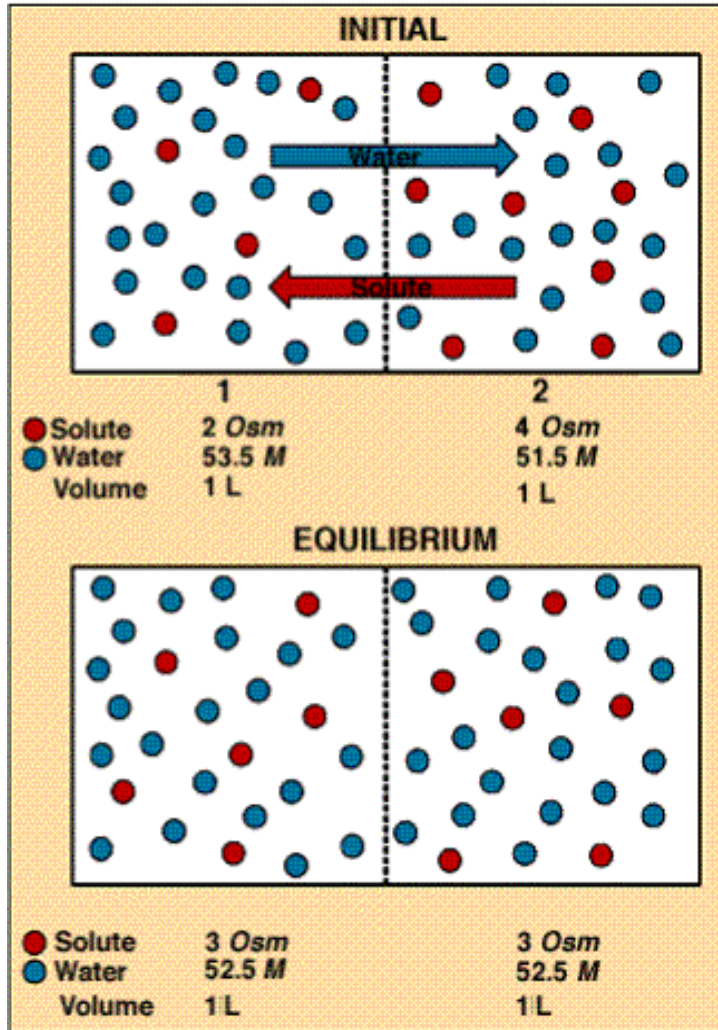
Each compartment has different electrolyte constituents kept in a defined range
 Osmolarities are equal but solute concentrations not in chemical equilibrium
 Homeostasis is not about reaching equilibrium but having parameters kept in a defined range by maintaining a steady state (energy required).

Steady State and Equilibrium

- Equilibrium and Steady State are NOT the same thing
- When a system is in balance or harmony, it is in a steady state
- **Steady State**: Maintenance of a state that does not change with time, and energy expenditure may be necessary
e.g. ECF $[\text{Na}^+] > \text{ICF } [\text{Na}^+]$ and concentrations are maintained at a set level balanced over time $\rightarrow$ requiring energy
- **Equilibrium**: Implies that two compartments have the same amount of free energy so there is no net energy transfer between two compartments
- In our body: ECF and ICF solutes are in a steady state (body expends effort and energy to maintain uneven solute concentrations to allow optimal function) but water is in equilibrium between the compartments (due to solute concentration difference)

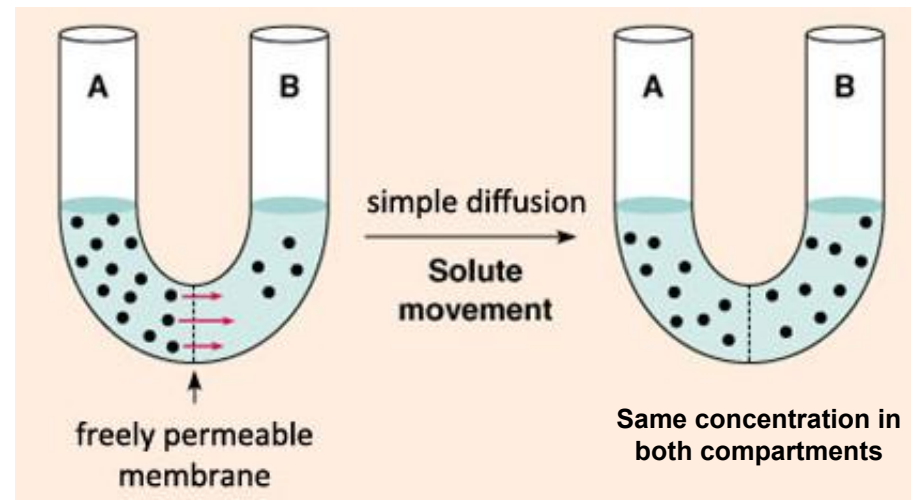
Equilibrium

- There is no net flux of energy from one compartment to another
- Rate of forward reaction = Rate of reverse reaction $\rightarrow C_1 = C_2$



At equilibrium

This reaction is in dynamic equilibrium because products are being converted to reactants and reactants are being converted to products.

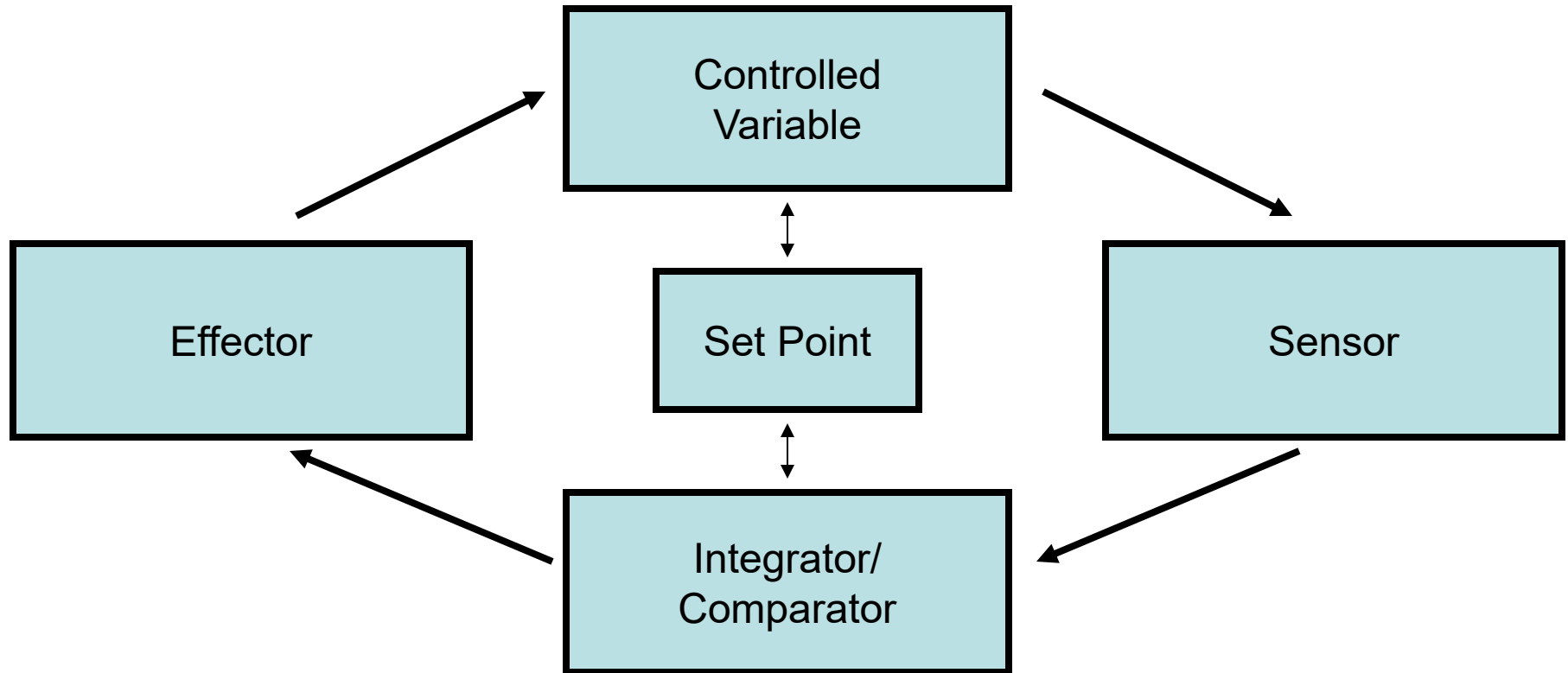


Steady State

- The tendency of a system to maintain internal stability, owing to the coordinated response of its parts to any situation or stimulus tending to disturb its normal condition or function
- Parameters do not change over time i.e. concentration of Na^+ in intracellular/extracellular fluids
- However, this is an energy requiring process
- Most homeostatic mechanisms require a great deal of energy to keep the system in steady state

Homeostatic Mechanisms

- Most homeostatic mechanisms operate through negative feedback



Controlled Variable: Physiological parameter being controlled

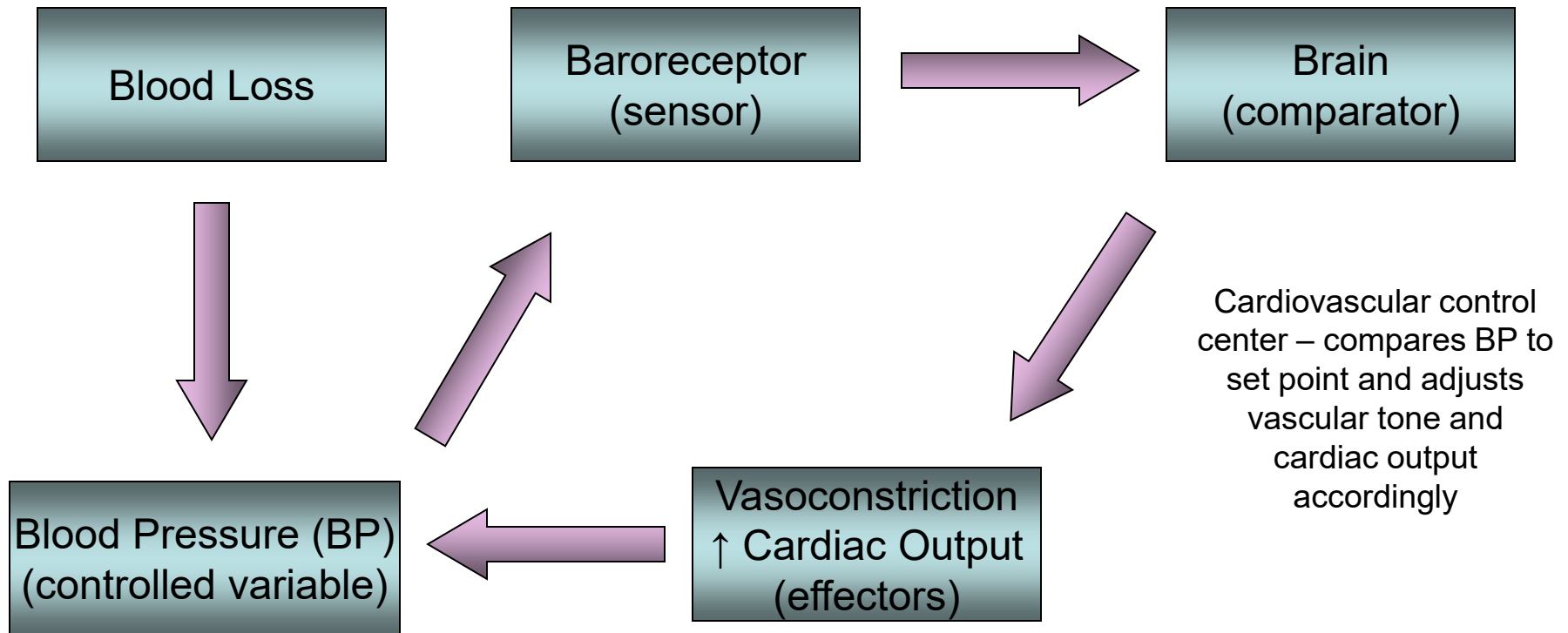
Sensor: Receptor type (sense organ) that detects changes in the controlled variable

Integrator/Comparator: Integration center that analyzes data from the sensor and compares to set point

Set Point: “Normal” values predetermined but influenced by environmental adaptations

Effectors: Nerve pathways, hormones, cells, tissues that carry out the response needed to restore homeostasis

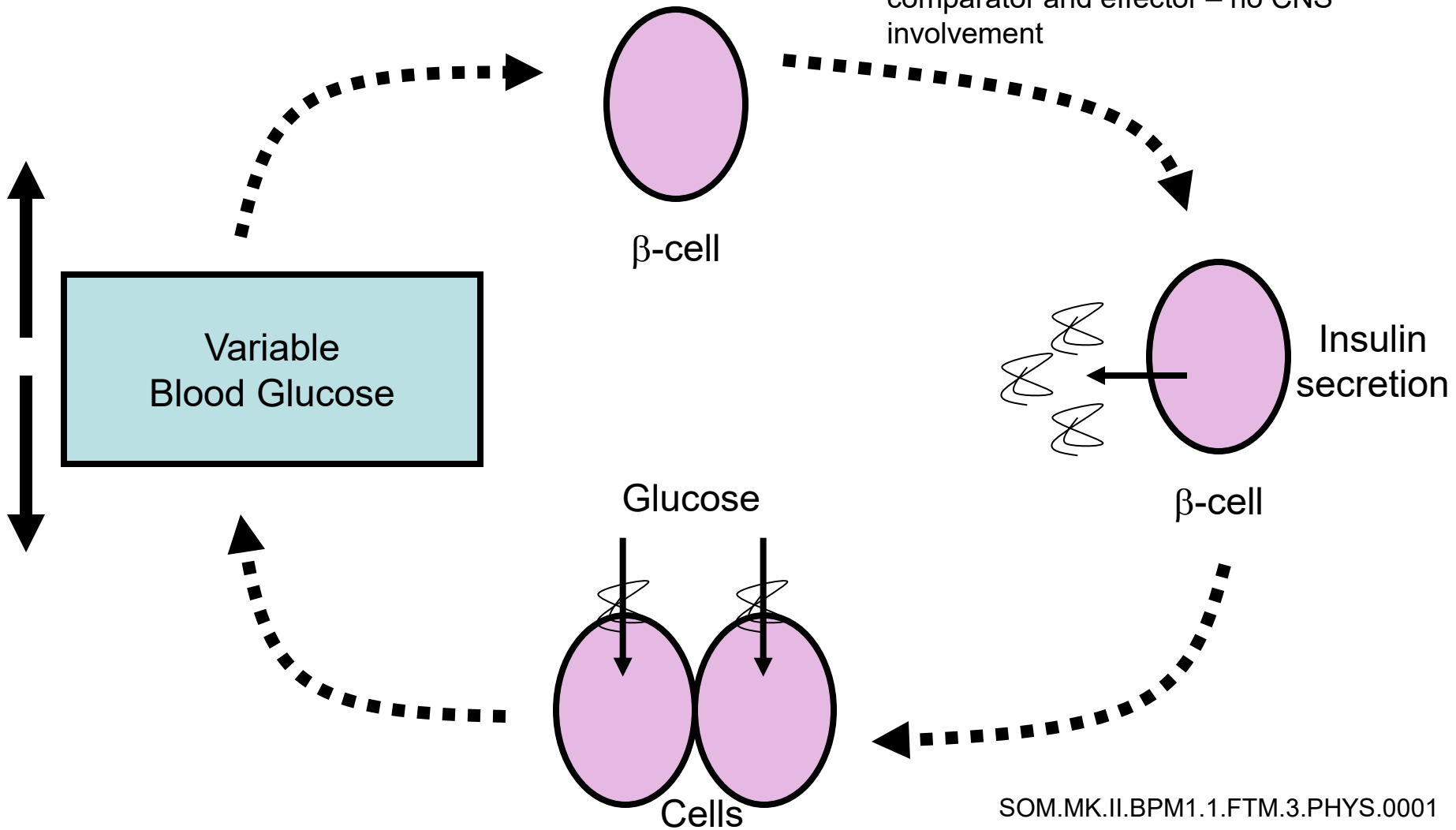
Blood Pressure Regulation: Hemorrhage



Blood loss → Hypovolemia → Decreased BP → Baroreceptors in carotid sinus and aorta sends info to CNS → CNS decides if BP is above/below set-point → restores BP back to normal by using autonomic nervous system which acts on various effector organs (heart and vasculature) to increase HR, contractility and decrease vascular diameter

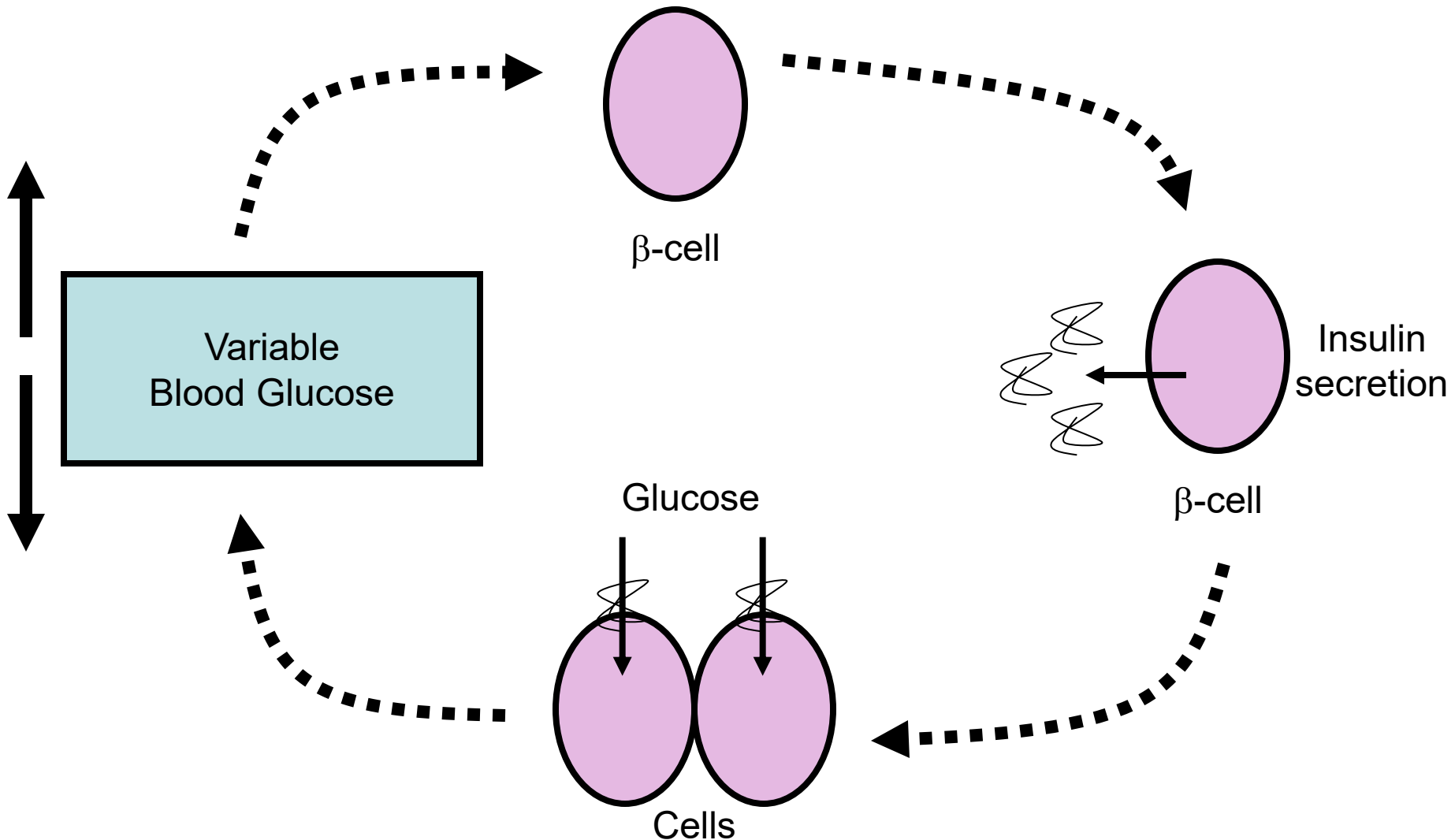
Blood Glucose Negative Feedback

Increase in blood glucose concentration will lead to pancreas secreting more insulin causing glucose uptake into cells and reducing plasma glucose concentration. Pancreas β -cells are the sensor, comparator and effector – no CNS involvement

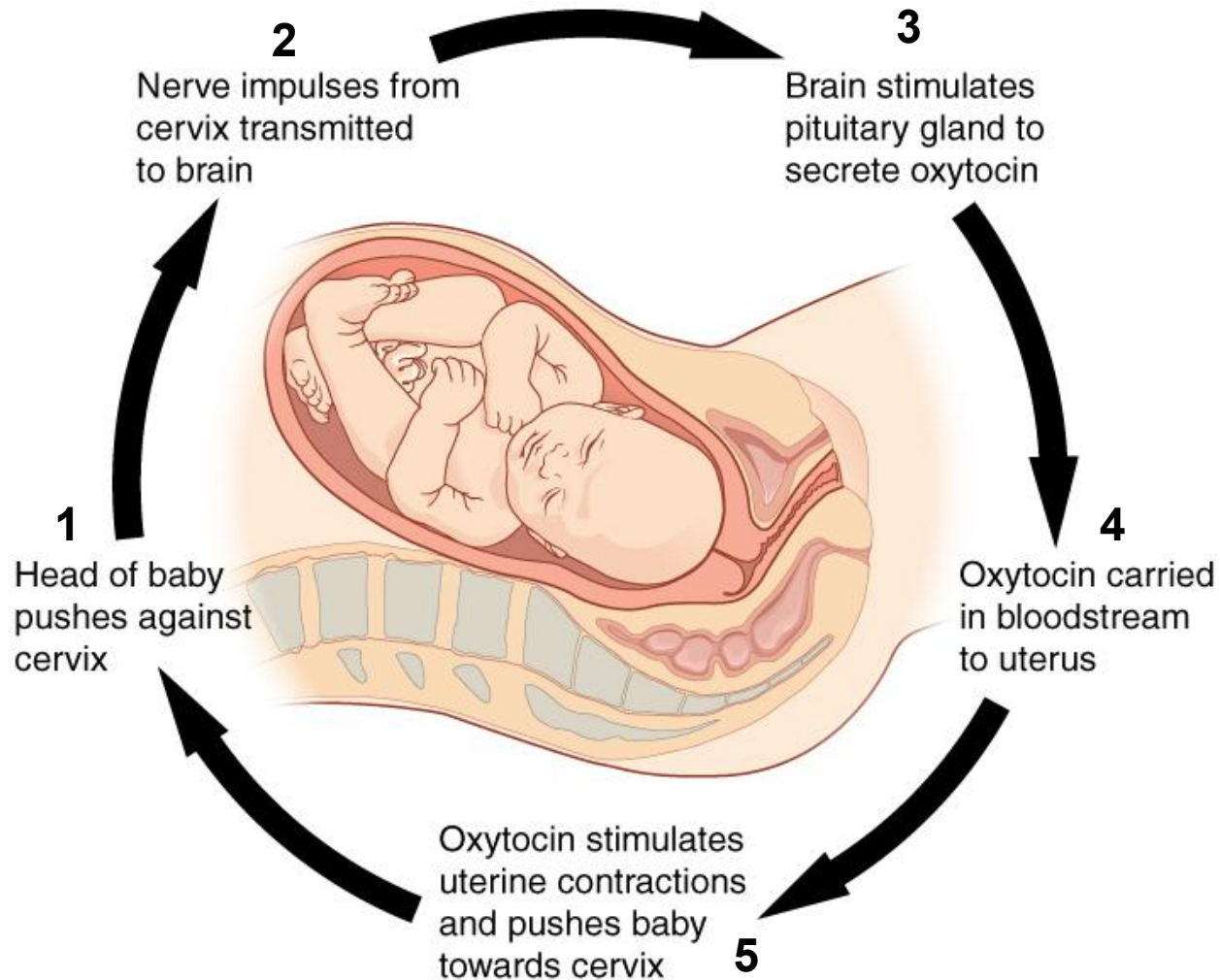


Blood Glucose Negative Feedback

Decrease in blood glucose concentration will lead to decreased insulin secretion maintaining higher concentrations. Pancreas β -cells are the sensor, comparator and effector.



Positive Feedback



Increase in uterine muscle tension leads to oxytocin release that causes an increase in uterine muscle contraction and more tension develops. These loop systems ultimately end in a definitive end point → birth

Feed-Forward Control

- Body anticipates a change in a variable before it happens and prepares the body for the change
 - a) Start line of a race: cardiac output and heart rate increase although no metabolic demand has been placed on body
 - b) Salivation that occurs upon smelling food – before food enters mouth salivation has increased to help lubricate and solvate food

Set Point Deviations

- Can vary individually or over time within the same person
- Set point may deviate due to:
 1. A 24hr Cycle: Circadian rhythm – since daily routine is so variable set points often vary between active (day) and passive (night) times
 - i.e. K^+ excretion is greater during the day than at night – more ingested during the day
 2. Environmental Changes: Changes in external environment
 - i.e. Acclimation → one goes to live at high altitudes, set point for PO_2 levels change

Set Point Deviations (Cont'd)

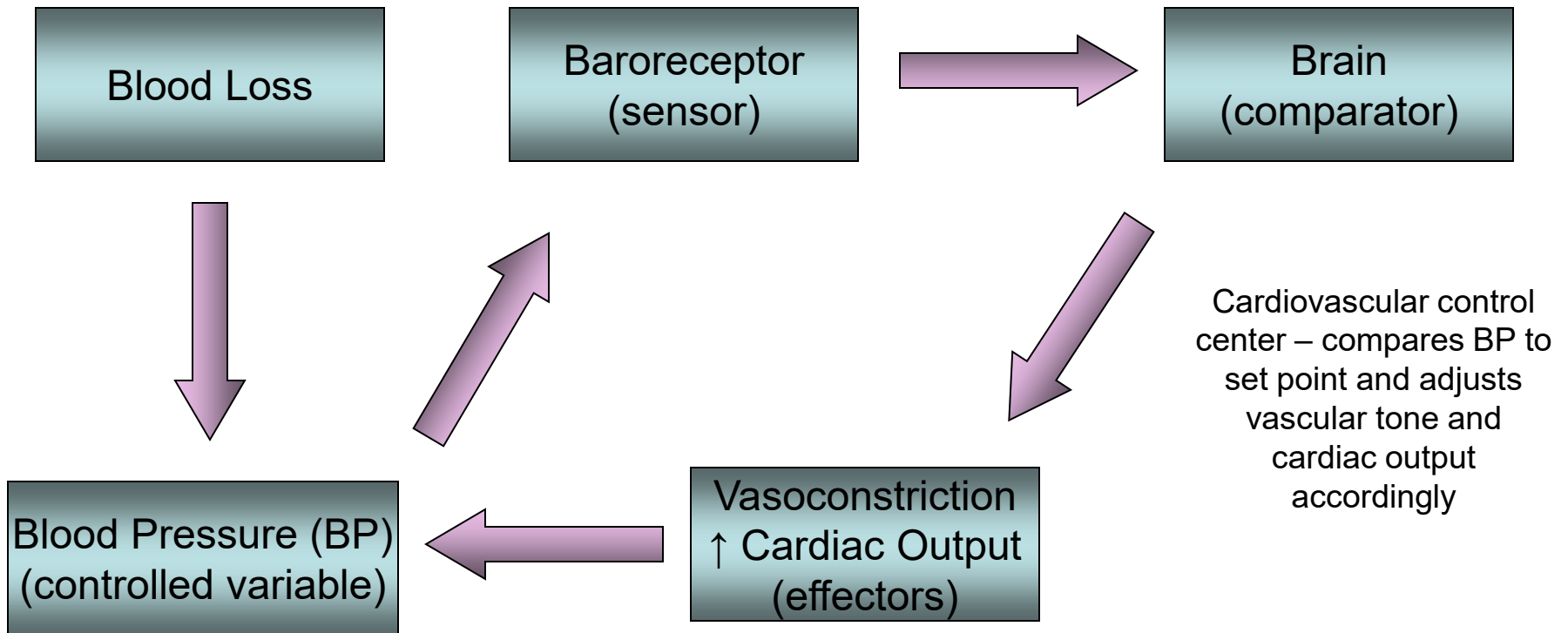
3. Protective Response: During infections
 - i.e. Fever - increase in body temperature is an adaptation of set-point to minimize viral replication
 - i.e. Plasma iron concentration decreases significantly to deprive bacteria of iron needed for replication

4. Ageing or Pathological changes: As disease progresses, set point elevates more and more
 - i.e. Atheromatous plaques in blood vessels increase BP and resetting of BP set point → blood pressure continues to increase chronically as cardiovascular disease progresses

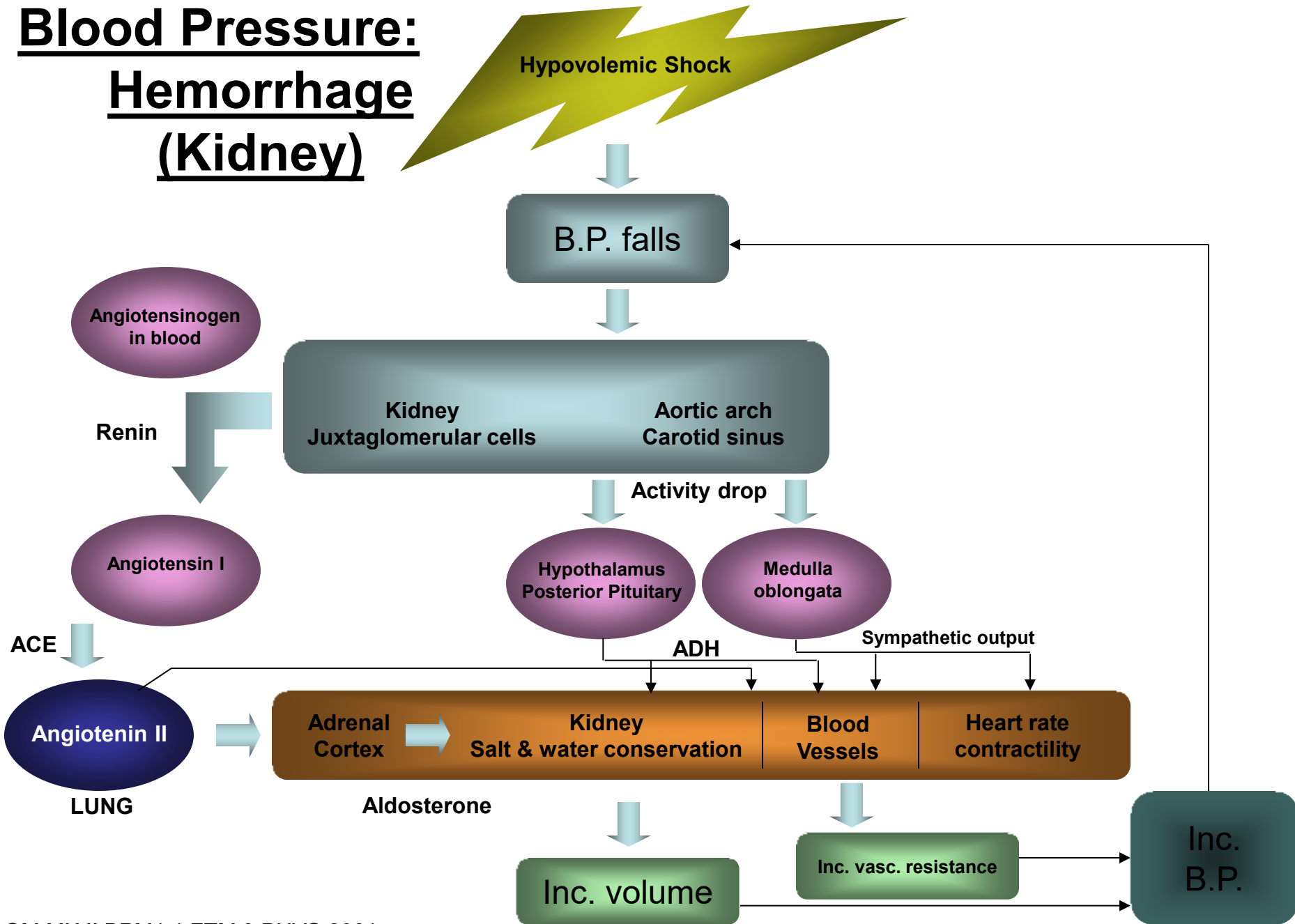
Redundancy

- The more vital the parameter to be controlled the more homeostatic mechanisms needed to control it
- Complex interplay may exist among various control systems
- If one mechanism fails then others can still maintain that parameter around set point → redundancy of control systems
- Often there is more than 1 control mechanism as a back-up

Blood Pressure: Hemorrhage (CV)



Blood Pressure: Hemorrhage (Kidney)

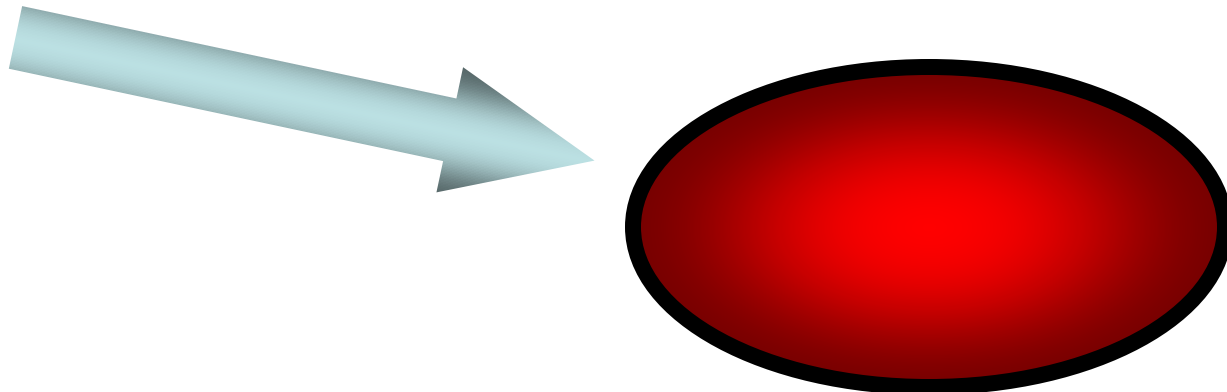


Homeostatic Hierarchy

Hierarchy of importance - where set-point of some variables may be altered in order to maintain others i.e. changes in skin blood vessel diameter during exercise

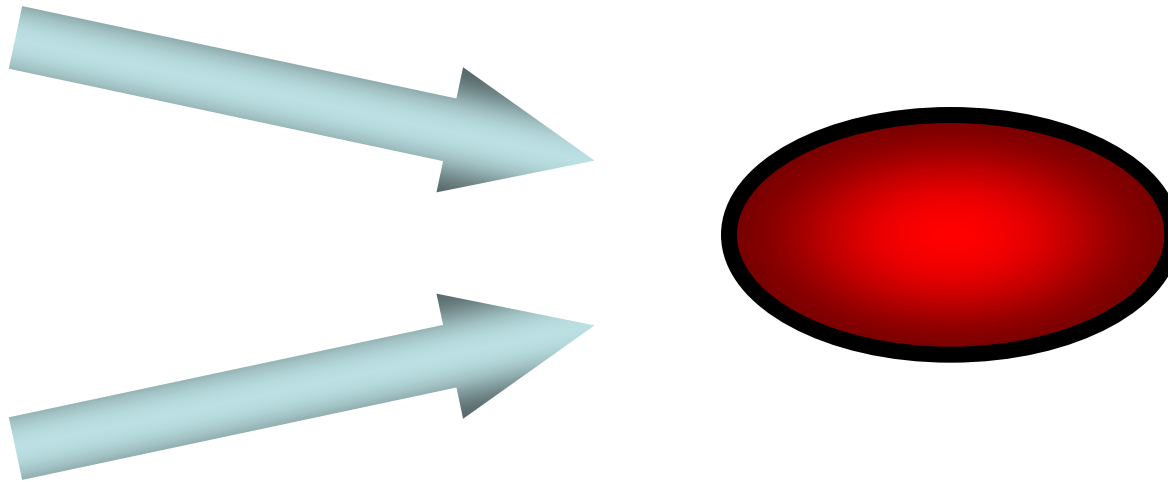
Cutaneous Blood Vessels

1. Prior to and during exercise
Sympathetic outflow increases causing
vasoconstriction
To help maintain blood pressure



Cutaneous Blood Vessels

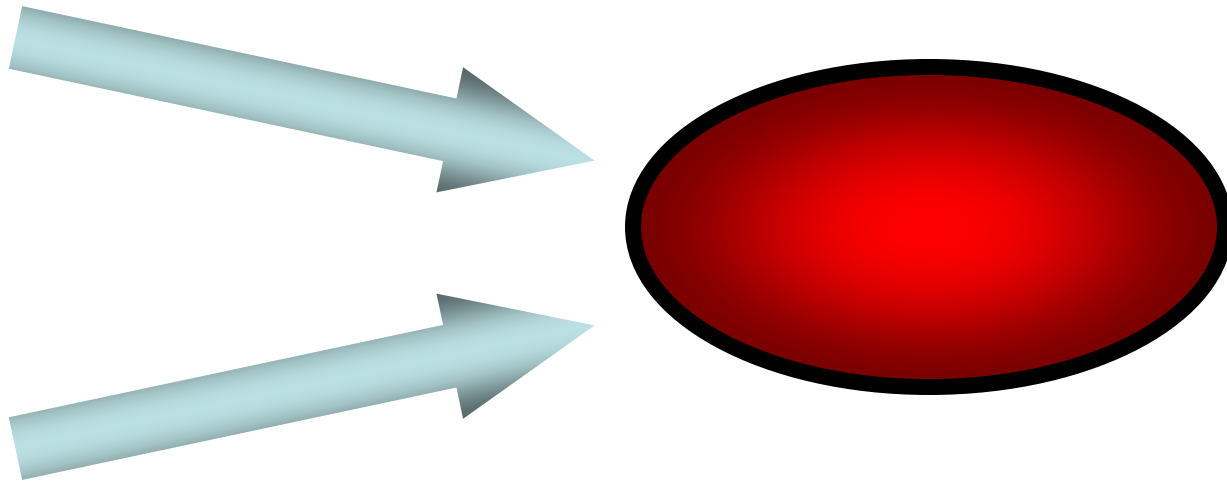
Prior to and during exercise
Sympathetic outflow increases causing
vasoconstriction
To help maintain blood pressure



2. With strenuous exercise, temperature control center in hypothalamus
detects heat increase and inhibits
Sympathetic outflow to skin blood vessels causing
vasodilation and heat loss

Cutaneous Blood Vessels

3. During maximal exercise the need to maintain Blood pressure takes priority over maintaining body temperature. Blood vessels constrict again



With strenuous exercise, temperature control center in hypothalamus detects heat increase and inhibits Sympathetic outflow to skin blood vessels, causing vasodilation and heat loss

Deficient Homeostatic Mechanisms

- Deficient homeostatic mechanisms lead to disease states
 - In many disease processes the normal negative feedback mechanisms may initially work to the benefit of the individual but as the disease progresses then normal compensatory responses fail and the normal negative feedback mechanisms are overtaken by positive feedback loops with disastrous consequences
- i.e. In chronic heart failure cardiac output falls and the body responds by retaining fluid. The retention of fluid increases the work load of the heart thus eventually making it even more likely to fail

Summary

- Homeostasis: Maintenance of a stable internal environment despite external variances
- Steady state: A state unchanging with time that needs energy expenditure may be necessary
- unchanging with time
- Equilibrium: When parameters are maintained with no net energy transfer
- Redundancy: More than 1 system to control a controlled variable (backup system)
- Hierarchy: Under certain circumstances systems compete to change a variable, each system has a priority

Summary (cont'd)

- Negative Feedback: Feedback causes a perturbation to be minimized or reversed to keep parameter at a set point
- Positive Feedback: Amplification of a deviation (usually has defined end point)
- Feed-Forward: Anticipation of and preparation for a change in a variable before it happens